GGG–O–591e

JUNE 5, 1962

SUPERSEDING
Int. Fed. Spec. GGG–O–0059ld (GSA–FSS)
FEBRUARY 11, 1962,
Fed. Spec. RR–O–376a
APRIL 30, 1953, and
Fed. Spec. RR–C–87
DECEMBER 19, 1938
(See 6.4)

FEDERAL SPECIFICATION

OILER, HAND

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers oilers of various types for use in ejecting oil under pressure or gravity or a combination of both (see 6.1).

1.1.1 *Federal specification coverage.* Federal specifications do not include all types, classes and styles of the commodities indicated by the titles of the specifications or which are commercially available, but are intended to cover the types, etc. which are generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types, classes, and styles.* The oilers covered by this specification shall be of the following types, classes, and styles as specified (see 6.2):

Type I.—Push-bottom, steel.
  Class A.—With machine-cut threads.
  Class B.—With rolled threads or other similar connection.

Type II.—Pump.
  Class A.—With handle and lever-operated pump.
  Class B.—Without handle.
  Class C.—With handle and plunger-operated pump.
    Style 2.—Bent spout.
    Style 3.—Curved spout.
    Style 4.—Flexible spout.

Type III.—Filler.

Type IV.—Rectangular, push sides.

Type V.—Typewriter (see 3.15).
  Style 1.—Oval shape.
  Style 2.—Cylindrical shape.
  Style 3.—Hemispherical shape.

2. APPLICABLE SPECIFICATIONS, STANDARDS AND OTHER PUBLICATIONS

2.1 **Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

VV–O–526—Oil; Lubricating, Machine.
PPP–B–566—Boxes, Folding, Paperboard.
PPP–B–576—Box, Wood Cleated, Veneer, Paper Overlaid.
PPP–B–585—Boxes, Wood, Wirebound.
PPP–B–591—Boxes, Fiberboard, Wood-Cleated.
PPP–B–601—Boxes, Wood, Cleated-Plywood.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636—Box, Fiberboard.
PPP–B–665—Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP–B–676—Boxes, Set-Up, Paperboard.
PPP–T–76—Tape, Pressure-Sensitive, Adhesive, Paper, Water-Resistant, for Carton Sealing.
PPP–T–97—Tape, Pressure Sensitive, Adhesive, Filament Reinforced.

FSC 4930

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Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods of.

MIL-L-10547—Liners, Case, Waterproof.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Bureau of Standards Handbook:

H-28—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations used herein are for the convenience of identification and are not intended to preclude the purchase of oilers which are otherwise in accordance with this specification.

3.2 Material. Material shall be as hereinafter specified. Materials not specifically designated shall be of good commercial quality. Contact between dissimilar metals, especially brass, copper, or steel in contact with aluminum or magnesium shall be avoided wherever possible.

3.3 Spouts. When specified (see 6.2), oilers shall be furnished with a flexible-metal spout and shall be so designed and constructed that the spout can be bent 90 degrees and remain in that position within 5 degrees.

3.4 Washers. Washers, if necessary for proper operation of the oilers, shall be furnished. The washers shall be impervious to oil and shall have adequate oil-sealing qualities.

3.5 Marking. Except for type V typewriter oilers, each oiler shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined. Type V oilers need not be marked. For Military procurement the oilers shall be marked in accordance with MIL-STD-130.

3.6 Finish. Type I, class A oilers shall be either polished or copper plated and finished with a coat of lacquer or varnish. Type II, class C oilers shall be finished by zinc plating, a coat of oil-resistant enamel, or coated with metal powder (aluminum, brass, or bronze) in an enamel oil base and baked. All other types and classes of oilers shall be polished, brass or copper coated, brass or copper plated, with a coat of clear lacquer or varnish, or lacquered with an opaque lacquer or nickel plated. When the basic metal used to fabricate oilers is corrosion resistant, no further coat-

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ing will be required. If lacquer, varnish, or enamel are used for the final coating they shall be oil resistant and shall not adversely affect the surface to which applied. All metal, regardless of type of treatment, shall be free of foreign matter which might adversely affect coating adhesion.

3.6.1 *Polished and clear-lacquered finish.* Polished and clear-lacquered finish, when examined under normal illumination without magnification, shall be a polished-surface finish free from drawing scratches or other objectionable marks. The polished surfaces shall be completely coated with a clear transparent lacquer. The lacquer shall not be tacky, shall not adversely affect the material coated, and shall be suitable for the prevention of rust.

3.6.2 *Polished, brass or copper coated, or brass or copper plated, and lacquer finish.* Polished, brass or copper coated, or brass or copper plated, and lacquered finish shall be a polished or otherwise suitably finished surface free from drawing scratches or other objectionable marks that might appear through the brass or copper plating or coating. Unless otherwise specified in the contract or order, the brass or copper plating or coating may be applied by an electro or other similar deposited method (see 6.2). The copper plating or coating shall have adequate corrosion-resistant properties. The brass or copper plating, unless applied by an electrostatic spray method shall be coated with a clear transparent lacquer. The lacquer shall not be tacky and it shall not adversely affect the copper plating.

3.7 Construction. The oilers shall be complete so that they can be used for the purpose intended.

3.7.1 *Castings.* If castings are used, they shall be of uniform quality, free from blowholes, porosity, or other defects which may adversely affect the strength or serviceability of the oiler.

3.7.2 *Welding, brazing, or soldering.* Welding, brazing, or soldering shall be in accordance with good commercial practice. In no event shall such processes, including plugging and peening, be used for reclaiming any defective parts of the oilers.

3.7.3 *Fastening devices.* If fastening devices are used, such as screws, bolts, nuts, pins, and similar parts, they shall be installed in a manner to prevent loss of tightness and adjustment. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.7.4 *MIL standard parts.* For Military purchases, MIL standard parts shall be used where applicable. In the event that MIL standard parts are not available, commercial utility parts, screws, nuts, bolts, etc., may be used provided they conform to the requirements of this specification.

3.8 Type I, class A, push-bottom steel oilers with machine-cut threads. Type I, class A oilers shall be similar to figure 1 and shall consist essentially of a body, washer as specified in 3.4, and a spout. The spout and body shall

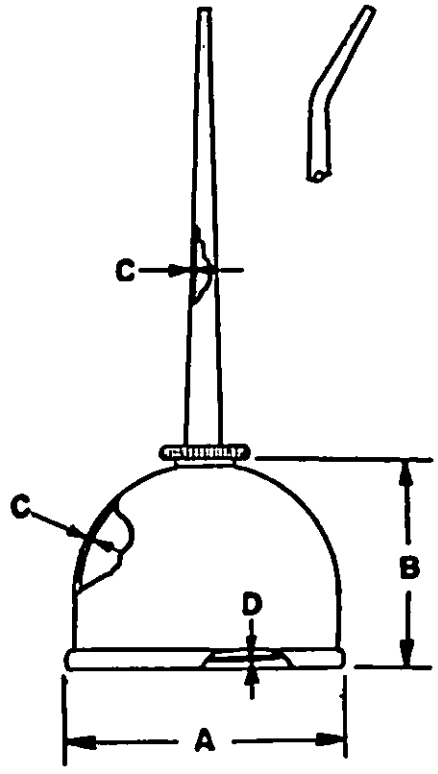


FIGURE 1. Type I, class A, push-bottom steel oiler with machine-cut threads.

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TABLE I. *Type I, class A, push-bottom steel oilers with machine cut threads*

Size, nominal	"A" diameter		"B" height		"C" wall thickness	"D" bottom thickness	Capacity
	Minimum	Maximum	Minimum	Maximum	Minimum	Minimum	Minimum
Pint	*Inches*	*Inches*	*Inches*	*Inches*	*Inch*	*Inch*	*Fluid ounces*
⅛	3	4	2	3	0.022	0.012	5
½	3	4	2	3½	.022	.012	8
¾	3	5	2	4	.022	.014	10
1	3½	5½	2	4	.022	.014	16

be connected by means of machine-cut threads in both parts. Material thicknesses and dimensions shall be in accordance with table I for the size specified (see 6.2).

3.8.1 *Body.* The body for type I, class A oilers shall consist of a one-piece, seamless, cold-rolled steel, drawn top section and a spring-action bottom. The drawn top section and bottom shall be securely fitted to each other and shall form a strong and leakproof joint. The body shall have an opening in the top of not less than ¾-inch diameter. This opening shall be either fitted with a solid-brass collar or the top part of the body opening shall be reinforced with a solid-metal neck in a manner to prevent weakening connections when the threads are cut.

3.8.2 *Spouts.* Type I, class A oiler spouts shall be of cold-rolled steel and shall be furnished in lengths of 4, 6, 9, or 12 inches as specified (see 6.2). If no spout length is specified, spouts 4 inches long shall be furnished for the ⅛- and ½-pint oilers and spouts 6 inches long for the ¾- and 1-pint oilers. Spouts shall be either straight or bent, as specified (see 6.2). The spout shall be fitted with a solid-brass bushing having machine-cut threads suitable for engagement in the threaded opening of the body. The spout seam shall be welded, brazed, or silver soldered. Lead-tin solders are not acceptable. The fitting of the collar and spout shall afford a strong and leakproof joint.

3.9 Type I, class B push-bottom steel oilers with rolled threads or other similar connection. Type I, class B oilers shall be similar to figure 2 and shall consist essentially of body, washer as specified in 3.4, and a spout. Unless otherwise specified (see 6.2), the body and spout shall be connected by bayonet type, rolled threads, or a similar connection on both

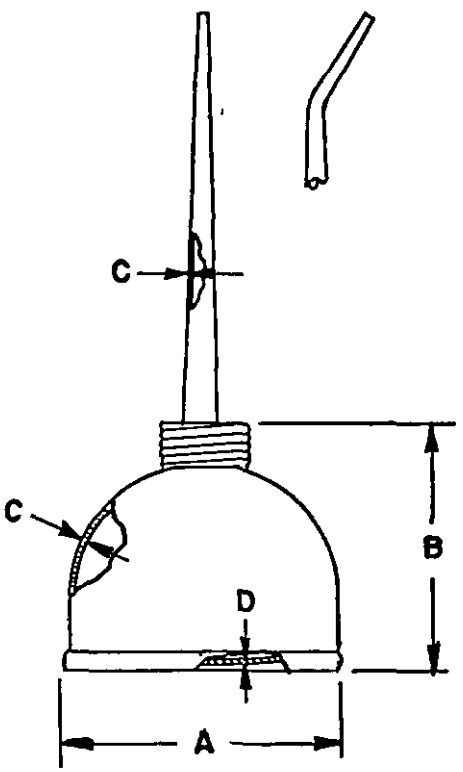


FIGURE 2. Type I, class B, push-bottom steel oiler with rolled threads or other similar connection.

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TABLE II. *Type I, class B, steel push-bottom oilers with rolled threads or other similar connection*

Size, nominal	"A" diameter		"B" height		"C" wall thickness	"D" bottom thickness	Capacity
	Minimum	Maximum	Minimum	Maximum	Minimum	Minimum	Minimum
	Inches	*Inches*	*Inches*	*Inches*	*Inch*	*Inch*	*Fluid ounces*
⅛ ounce	1⅛	2⅛	1	2	0.012	0.010	⅛
1 ounce	1⅛	2⅛	1	2	.015	.010	1
1½ ounce	2	3	1	2½	.015	.010	1½
½ pint	3	4	2	3	.017	.010	5
½ pint	3	4	2	3½	.017	.010	8
¾ pint	3	5	2	4	.018	.010	10
1 pint	3½	5½	2	4	.018	.010	16

parts. The oilers shall be in accordance with table II for the size specified (see 6.2).

3.9.1 *Body*. The body of type I, class B oilers shall consist of a one-piece seamless cold-rolled steel top section and a spring-action bottom. The top section and bottom shall be securely fitted to each other with a strong and leakproof joint. Oilers of 1½-ounce capacity and smaller shall have an opening in the top of the body of not less than ½-inch diameter and oilers larger than 1½-ounce capacity shall have an opening of not less than ¾-inch diameter. Rolled threads or other similar connection shall be provided at the top of the body to receive the spout.

3.9.2 *Spouts*. Spouts for type I, class B oilers shall be of cold-rolled steel and shall be 1⅞ to 4⅜ inches long for oilers of 1½-ounce capacity and less, and in lengths of 3, 5, or 9 inches for oilers larger than 1½-ounce capacity, as specified (see 6.2). If no spout length is specified, spouts 5 inches long shall be furnished for ¾- and 1-pint oilers and spouts 3 inches long for smaller-sized oilers. Spouts shall be either straight or bent, as specified (see 6.2). The spout shall be fitted with a cold-rolled steel cap having bayonet type, rolled threads or other similar connection to fit the mating connection on the body. The spout seam shall be welded, silver soldered, or brazed. Lead-tin solders are not acceptable. The fitting of the cap and spout shall afford a strong and leakproof joint.

3.10 Type II, class A pump oilers with handle and lever-operated pump. Type II, class A oilers shall be similar to figure 3 and shall consist essentially of a body, a washer as specified in 3.4, handle, spout, and pump. The body, handle, and spout shall be of cold-rolled steel. The pump shall be easily removable from the can. The oilers shall comply with the requirements of table III for the size specified (see 6.2).

3.10.1 *Body*. Bodies shall be provided with a top section either of drawn or fabricated seam construction having welded, brazed, or silver soldered seams and shall be not less than 0.018 inch thick for the 1- and 1½-pint size and not less than 0.025 inch thick for the 2-pint size. Lead-tin soldered seams will be acceptable provided the soldered seams are at least 1 inch from the bottom of the can. The bottom shall be securely fitted to the top section to afford a strong and leakproof joint. The body shall have an opening of the top not less than 1-inch diameter with threads or other similar connection for receiving the pump unit. Bottom shall be connected to the top section by double-locked seam, or of drawn construction provided it extends at least one inch up the side of the can.

3.10.2 *Pump*. The pump shall be self-prim-

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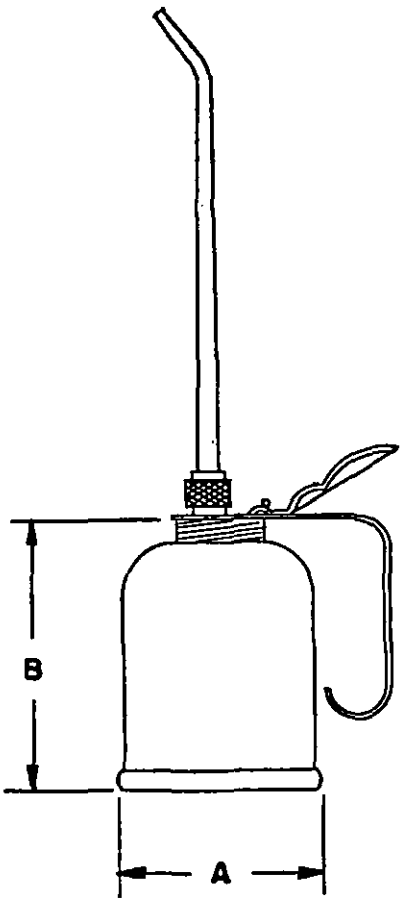


FIGURE 3. Type II, class A, pump oiler with handle and lever-operated pump.

TABLE III. *Type II, class A, pump oilers with handle and lever-operated pump*

Size nominal	"A" diameter		"B" height		Capacity
	Minimum	Maximum	Minimum	Maximum	Minimum
Pints	*Inches*	*Inches*	*Inches*	*Inches*	*Fluid ounces*
¾.....	3	4	3	5	10
1.....	3	4	3	5	16
1½.....	3	4½	4	6	24
2.....	3½	5	5	7	32
4.....	4	6	6	8	60

ing and shall function in any position with 8, 10, 20, or 30 grades of oil or any similar grades that will flow and are ordinarily used for lubricating purposes. The pump parts shall be of metal construction except that the lower pump tube may be of nylon (see 4.4.2). The pump shall be operated by a conveniently located lever. The pump shall be capable of removing at least three-fourths of the oil contents of the can without its being tilted from the normal upright position.

3.10.3 *Handle.* The handle stock thickness shall be not less than 0.032 inch and the edges shall be rolled back at least 180 degrees, or otherwise suitably reinforced, or shall be made of flat strip, with rounded edges, not less than 0.062 inch thick.

3.10.4 *Spout.* Spouts shall be 6, 7, 9, 12, or 15 inches long as specified (see 6.2). If no spout length is specified, the 6-inch spout shall be furnished. Rigid spouts shall be seamless or have a welded, silver soldered or brazed, leakproof seam and shall be not less than 0.017 inch thick. Tin-lead solders will not be acceptable. The spout shall be readily detachable and joined to the pump unit by bayonet-type joint, a threaded joint, or other suitable connection. Spouts shall be either straight, bent at the vent end, or angled at the base and bent at the vent end, as specified (see 6.2). The bend or the angle and bend shall be in the opposite direction from the handle for offset filling.

3.11 Type II, class B pump oilers without handle. Type II, class B oilers (known commercially as pistol-grip oilers) shall be similar to the optional designs shown in figure 4 and shall be suitable for delivering either a drop or stream of oil, as wanted. The oiler shall be designed and constructed to operate satisfactorily at any temperature from minus 40° to plus 125° F. and withstand storage temperatures between minus 80° to plus 160° F. (see 4.4.1.3). The body shall afford a comfortable handgrip. These oilers shall consist essentially of a body, a washer as specified in 3.4, spout, and pump. The body and spout shall be of all metal construction. The oilers shall be in accordance with table IV for the size specified (see 6.2).

3.11.1 *Body.* The top section of the body shall be either one piece or of fabricated con-

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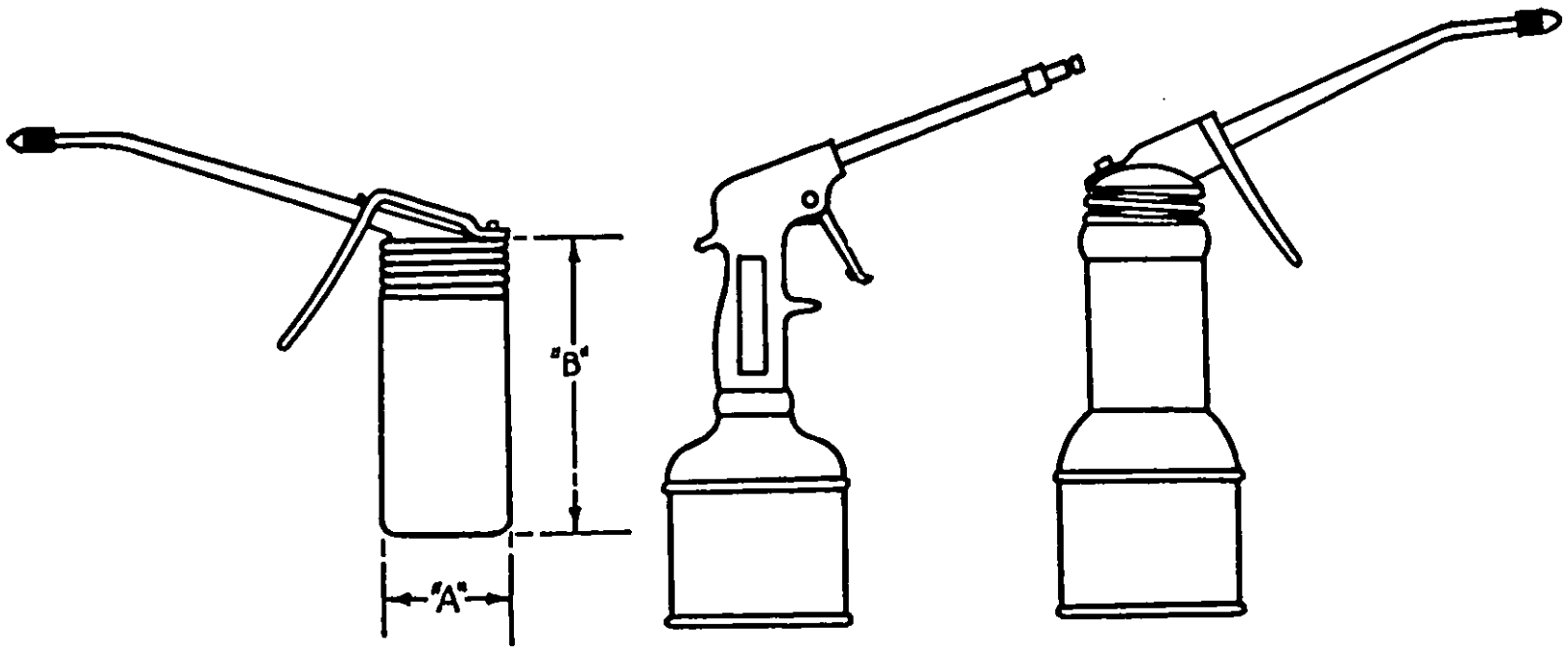


FIGURE 4. Type II, class B, pump oiler, without handle. (Optional designs, see 3.1).

TABLE IV. *Type II, class B, pump oilers without handle*

Size, nominal	"A" diameter (see note)		"B" height		Capacity	
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
Pints	*Inches*	*Inches*	*Inches*	*Inches*	*Fluid ounces*	*Fluid ounces*
½	1½	2½	3½	5¼	4	6
½	1½	2	4½	8	6	10
1	1½	2½	5½	10	16	22
2	1½	2½	8	11	28	36

Note: The maximum diameter applies to the hand grip area. A larger base diameter may be furnished if the base is of the flared design (see optional designs in figure 4).

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struction. Welding, silver soldering, or brazing of seams is permissible but lead soldering is not permissible unless the soldered seam is at least 1 inch from the bottom of the can. The metal in the top section and the bottom of the body shall be at least 0.015 inch thick.

3.11.2 *Pump.* The pump shall be self-priming and shall function properly at a temperature of minus 40° to plus 125° F. The pump shall function properly in any position with any oil of 8 or 10 grade that will flow readily and is ordinarily used for lubricating purposes. The structural parts of the pump shall be of metal except that the lower pump tube may be of nylon (see 4.4.2). Unless otherwise specified in the contract or order, the pump shall be operated by a conveniently placed lever or a plunger with button head. The pump shall be capable of removing at least 90 percent of the oil contents of the can without its being tilted from the normal upright position.

3.11.3 *Spout.* The spout shall be seamless or shall have a welded, silver soldered, or brazed seam and shall be not less than 0.020 inch thick. Unless otherwise specified in the contract or order, the outer end of the spout shall be fitted with a suitable cone-shaped nozzle. The length of the spout, including the nozzle fitting, shall be not less than 3 inches for ¼-pint size, 4 inches for ½-pint size and 6 inches for 1-pint size and larger. Unless otherwise specified in the contract or order, the spout shall be either bent or straight. When specified (see 6.2), the oiler shall be furnished with an adapter and spray nozzle.

3.12 Type II, class C pump oilers with handle and plunger-operated pump. Type II, class C oilers shall be similar to figure 5 and shall consist essentially of a body, top cap, bottom cap or spout, piston, piston rod, handle, and spout. When assembled and filled with oil, the pump shall be capable of withstanding a pressure of 25 pounds applied to the pump handle without leakage at joints or past the piston. Oilers shall conform to table V for the size specified (see 6.2). They shall be designed to supply lubricants or similar liquids by force feed to gear cases, shock absorbers, track suspension systems, brake fluid master cylinders, etc., and may be used as a suction pump. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment and repair.

TABLE V. *Type II, class C, pump oilers with handle and plunger-operated pump*

Gun (length) (minimum)	Capacity (minimum)	
Inches	*Pints*	*Ounces*
10		6.0
15		11.0
18	1.9	25.8

3.12.1 *Body.* The body shall be cylindrical in shape, not less than 1½ inches or more than 2½-inch outside diameter, of the length and capacity shown in table V, and as specified (see 6.2). The handle end of the body shall be provided with screw-type cap to permit the withdrawal of the piston for disassembly. The other end of the body shall be provided with a screw-type or welded-on-type cap, or swaged body provided with female pipe threads for attachment of the spouts. The wall of the body shall be not less than 0.0475 inch in thickness. The internal surface of the body shall be smooth to minimize wear on the "O" ring or piston caps. The outside central portion of the body shall be either continuously or interruptedly knurled for a minimum of 3½ inches in length.

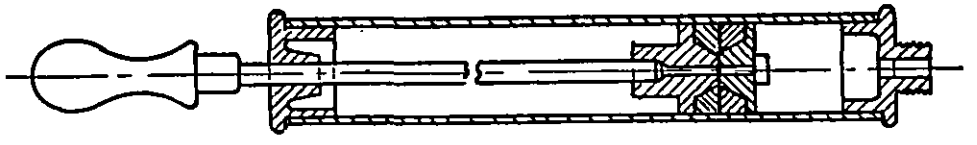


FIGURE 5. Type II, class C, pump oiler with handle and plunger-operated pump.

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3.12.2 *Caps.* Caps shall be threaded to fit the body, and arranged to provide leakproof joints. They shall be knurled on the outside periphery. A vent hole shall be provided in the cap at the piston end of the oiler if required for proper operation of the oiler. If a vent hole is used, it shall be approximately ⅛ inch in diameter and the piston hole shall be suitably reinforced with a projection or by other suitable method. The bottom caps shall have a spout connection of either ⅜ inch—18 NPSC external threads or ¼ inch—18 NPT internal threads, as specified (see 6.2), in accordance with the applicable requirements of National Bureau of Standards Handbook H28.

3.12.3 *Piston and piston rod.* The piston shall be of the flexible rubber "O" ring or cup type, fabricated from material that resists deterioration when subjected to prolonged contact with petroleum base and synthetic oils. The seals and sealing cups shall be réplaceable and shall be so constructed as to seal against leakage at 25 pounds per square inch pressure of No. 10 machine oil conforming to VV-O-526; the piston-rod diameter shall be not less than 0.300 inch.

3.12.4 *Handle.* A handle suitable in all respects for operating the piston shall be provided. The handle shall be smooth-finished hardwood, metal, or plastic. The handle shall be as shown in figure 5 or of the type in which the centerline of the piston rod intersects the centerline of the handle at its midpoint, and at right angles to it (including type known commercially as spade type), or of the solid pistol-grip type. When the piston is at the end of the discharge stroke, the clearance between the inner grip surface of the handle and the body shall be not less than ⅞ inch.

3.12.5 *Gaskets.* Gaskets of cork or other suitable material shall be provided for the end caps and spout connection, where required, to prevent leakage under any test and operating conditions.

3.12.6 *Spouts.* Unless otherwise indicated in the invitation for bids (see 1.2 and 6.2), one spout of the style specified in the invitation for bids shall be furnished with each oiler. One end of each spout shall be threaded to fit the discharge end cap of the oiler.

3.12.6.1 *Style 2.* Style 2, bent spout shall be made of welded or seamless tubing. It shall either be bent at right angle (each leg of angle being not less than 6 inches in length) or in the form of a gooseneck with an overall length of not less than 11 inches. The outside diameter of the spout shall be not less than ¼ inch. The inside diameter of the spout shall be not less than 3/16 inch (see figure 6).

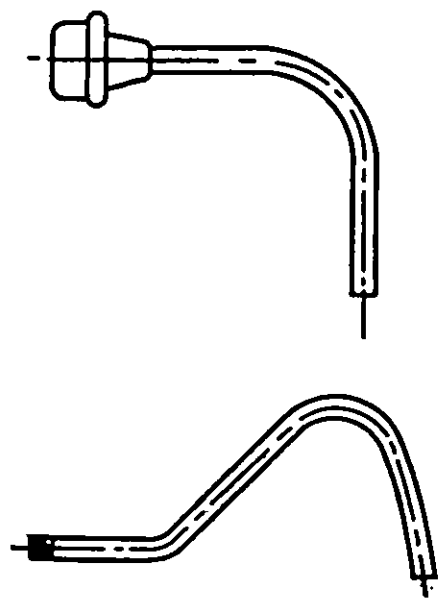


FIGURE 6. Type II, class C, style 2, bent spout for pump oilers with handle and plunger-operated pump.

3.12.6.2 *Style 3, curved.* Style 3, curved spout shall be made of welded or seamless tubing. It shall be curved on a 1½-inch-minimum radius so that the curved end is offset from the centerline of the piston at its extremity and shall be not less than 3⅝ inches in length with tip attached (see figure 7). A tip or nozzle shall be fitted at the outer end of the spout. The tip shall be either tapered to approximately ¼-inch diameter at the small end or it shall be straight and not more than ⅝-inch-outside diameter. The inside diameter of the tip shall be not less than 3/16 inch.

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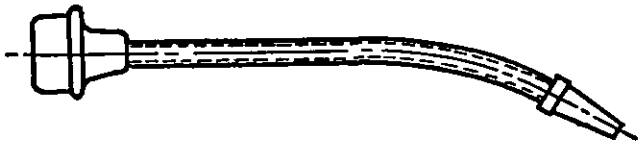


FIGURE 7. Type II, class C, style 3, curved spout for pump oilers, with handle and plunger-operated pump.

3.12.6.3 *Style 4, flexible.* Style 4, flexible spout shall be made of flexible-steel tubing. It shall be not less than 12 inches in length, and shall be suitable in all respects for reaching locations that would otherwise be inaccessible (see figure 8). Each end of the tubing shall be provided with a fitting securely fastened in place. The fitting on one end shall be threaded for attachment to discharge end cap of the oiler. The fitting on the other end shall be either a tapered nozzle or a straight ferrule. The outside diameter of the spout shall be not less than ¼ inch and the inside diameter shall be not less than 3/16 inch.

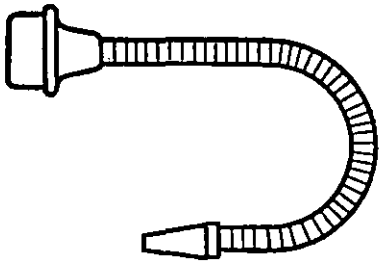


FIGURE 8. Type II, class C, style 4, flexible spout for pump oilers, with handle and plunger-operated pump.

3.12.7 *Temperature.* When specified (see 6.2), the oilers shall be so designed as to function between minus 40° to plus 125° F. and withstand storage temperatures between minus 80° to plus 160° F. without harmful effects (see 4.4.1.3).

3.12.8 *Filling.* The cylinder of the oiler shall be capable of being filled to within 85 percent of the capacity with lubricating oil by a single stroke of the piston.

3.13 **Type III filler.** Type III filler shall be similar to figure 9 and shall be made of either cold-rolled steel or copper, as specified (see 6.2), and shall consist essentially of a one-piece seamless body with attached bottom, handle, spout, cap, and chain. Fillers shall comply with the requirements of table VI for the size specified (see 6.2).

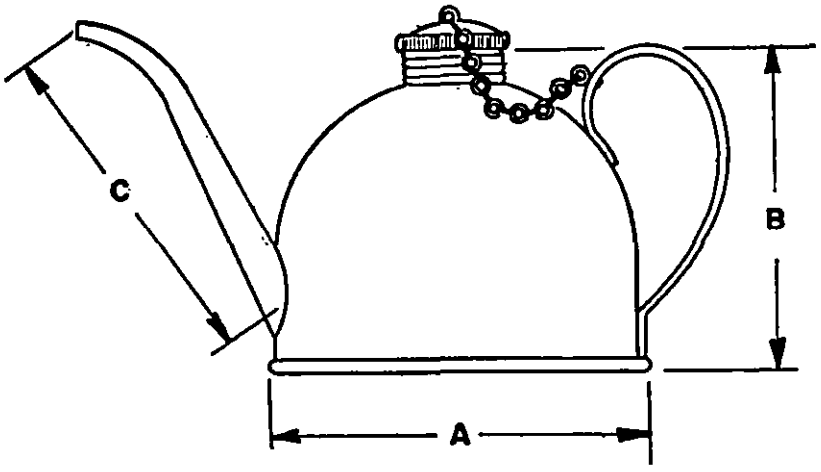


FIGURE 9. Type III, filler.

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TABLE VI. *Type III, fillers*

Size, nominal	"A" diameter		"B" height		"C" spout length		Capacity
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	Minimum
Pints	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Inches*	*Fluid ounces*
1	4	5	3	4	4	6	14
2	4 1/4	6	4	5 1/4	5 1/4	8	30
4	5 1/4	7	5	6 1/4	5 1/4	8	56

3.13.1 *Body.* The metal composing the top section and bottom of the body shall be not less than 0.025 inch thick. The top and bottom shall be joined securely to afford a strong and leakproof construction. A bushing shall be provided in the top of the body to be threaded for securing the cap.

3.13.2 *Spout.* The joint between the spout and body and all seams shall be welded, silver soldered, or brazed. Tin-lead solders will not be acceptable. Spout material shall be not less than 0.031 inch thick. The orifice at the outer end of the spout shall be not less than 1/4 inch in diameter. Unless otherwise specified in the contract or order, the spout shall be bent.

3.13.3 *Cap and chain.* The cap shall be made of either steel or brass bayonet-type joint, rolled threads, or machine-cut threads for properly engaging the bushing in the body. The cap shall be attached to the handle by means of a chain.

3.14 Type IV, rectangular push sides. Type IV oilers shall be similar to figure 10 and shall consist essentially of a rectangular body, two washers as specified in 3.4, and a spout. The body and spout shall be of cold-rolled steel not less than 0.020 inch thick. The design shall be such that the spout may be attached in the conventional manner for dispensing oil or inverted and screwed into the body for storage. The oiler shall be leakproof regardless of the position of the spout. The oilers shall comply with the requirements of table VII.

3.14.1 *Body.* Portions of two of the vertical sides of the body shall be convex in a manner to eject oil from the oiler when pushed and to return to normal position when pressure is released. The opening in the top of the body shall be not less than 3/4 inch in diameter and shall be provided with an all metal bushing that is threaded to receive the threaded portion of the spout. Soldered seams are permissible in this type oiler.

3.14.2 *Spout.* The orifice at the outer end of the spout shall be not less than 1/8-inch diameter. The small end of the spout shall be provided with a cap attached by a leakproof threaded connection and be readily detachable. The large end of the spout shall be provided with an externally threaded sleeve having a knurled retainer containing two washers (see 3.4), one on each side of the knurled portion. The knurled retainer shall be located midway between the two threaded ends of the sleeve so that either end of the sleeve may be screwed into the can opening. The spout shall be straight.

3.15 Type V typewriter oilers. Unless otherwise specified (see 6.2), the typewriter oilers shall be any of the following styles: Style 1, oval shape (see 3.15.1 and figure 11), style 2, cylindrical shape (see 3.15.2 and figure 12), or style 3, hemispherical shape (see 3.15.3 and figure 13. The oilers shall be capable of discharging oil, as applicable to the style (s) specified.

3.15.1 *Style 1.* Type V, style 1, oval shape, typewriter oilers shall be similar to figure 11 and shall consist essentially of a body, a washer as specified in 3.4, a spout, and a spout cover. The metal shall be not less than 0.008

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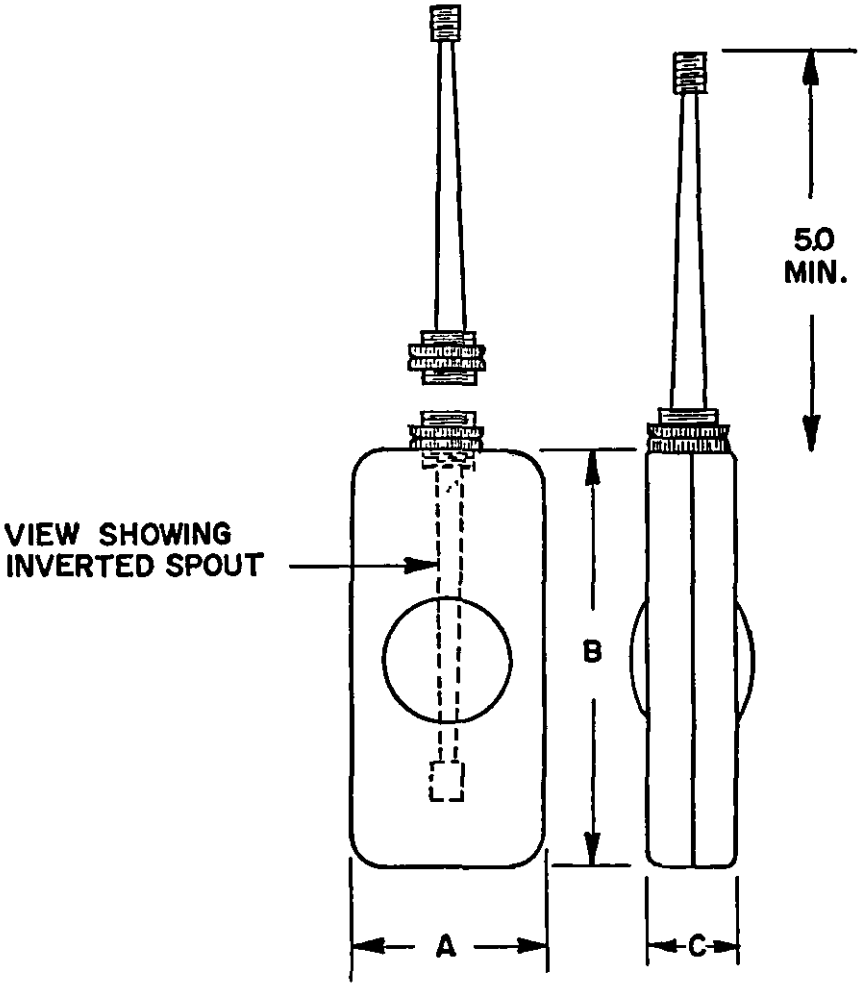


FIGURE 10. Type IV, rectangular, push sides oiler.

TABLE VII. *Type IV, rectangular, push sides*

Size, nominal	"A" width	"B" height	"C" thickness	Capacity
	Minimum	Minimum	Minimum	Minimum
Pint	*Inches*	*Inches*	*Inches*	*Fluid ounces*
¾.....		6½	1¼	12

inch in thickness. The oiler shall be capable of discharging one drop of typewriter oil through the tip end of the spout by tilting or pressing the sides of the oiler with the fingers. It shall be capable of discharging several drops of oil through the tip of the spout by tilting the oiler and gently pressing the sides of the body with the fingers. These oilers shall comply with the requirements of table VIII.

3.15.1.1 *Body.* Portions of two of the vertical sides of the body shall be convex in a manner to eject oil from the oiler when pushed and to return to normal position when pressure is released. Failure to return to normal position will be considered as failure to meet specification requirements. The opening in the top of the body shall be not less than ⅜ inch in diameter and shall be provided with an all metal collar that is threaded to receive the threaded

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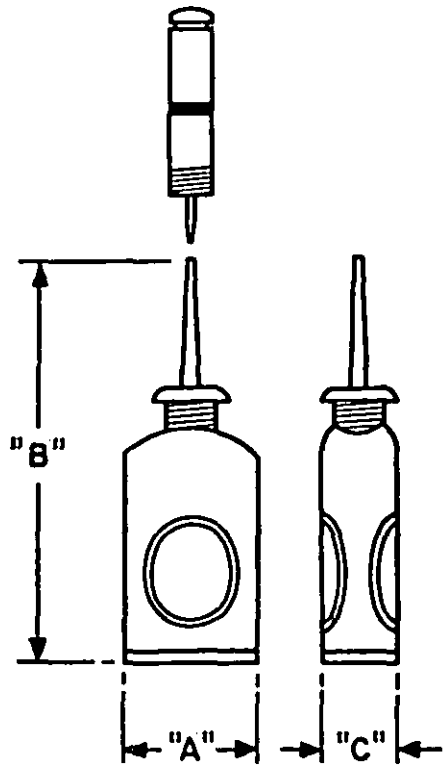


FIGURE 11. Type V, style 1, oval shape, typewriter oiler.

TABLE VIII. *Type V, style 1, oval shape, typewriter oiler*

Size, nominal	"A" width	"B" height	"C" thickness	Capacity
	Minimum	Minimum	Minimum	Minimum
Pint	*Inch*	*Inches*	*Inch*	*Fluid ounces*
½.....	1	3⅝	⅝	½±⅛

portion of the spout and the threaded portion of the spout cover. All seams shall be soldered to make them leakproof.

3.15.1.2 *Spout.* The orifice at the outer end of the spout shall be not less than 1/16 inch in diameter. The spout shall be not less than 1 inch long. The large end of the spout shall be threaded for insertion in the threaded portion of the body. The spout shall be straight.

3.15.1.3 *Spout cover.* The spout shall be provided with a cap or cover which shall prevent leakage, when in place. Attached to the inside of the cover shall be a pointed-spout cleaning wire so arranged that when the cover is screwed in place the cleaning wire shall extend inside the spout, or a neoprene plug shall be inserted in the inside of the extreme top of the cover, so located that when the cover is screwed into place the spout will press into the neoprene plug insuring a tight fit and prevent leaking. The open end of the cover shall be threaded for insertion in the threaded portion of the body. The spout cover shall have a knurled ring around it to provide a good gripping surface when inserting or removing the cover from the body of the oiler.

3.15.2 *Style 2.* Type V, style 2, cylindrical shape, typewriter oilers shall be similar to figure 12 and shall consist essentially of a body, a washer as specified in 3.4, a spout, and a spout cap. The metal shall be not less than 0.010 inch in thickness. The oiler shall be capable of discharging either one drop or several drops of oil through the tip of the spout by gently pressing the bottom of the oiler. These oilers shall comply with the requirements of table IX.

3.15.2.1 *Body.* The body of the oiler shall be cylindrical in shape and at the option of the supplier may be either straight or stepped as shown in figure 13. The opening in the top of the body shall be not less than ¼ inch in diameter. The top of the body shall be threaded for assembly of the spout. The bottom shall be of the push-bottom type. The bottom, when pressed, shall return to normal position when pressure is released. Failure of the bottom to return to normal position will be cause for rejection. Seams shall be leakproof. Nonmetallic filling of seams is not acceptable.

3.15.2.2 *Spout.* The orifice at the outer end of the spout shall be not less than 0.025 inch in diameter. The spout shall be not less than 1¾ inches long. The large end of the spout shall be threaded for assembly on the body. The tip or small end of the spout shall be threaded for assembly of the spout cap. The spout shall be straight.

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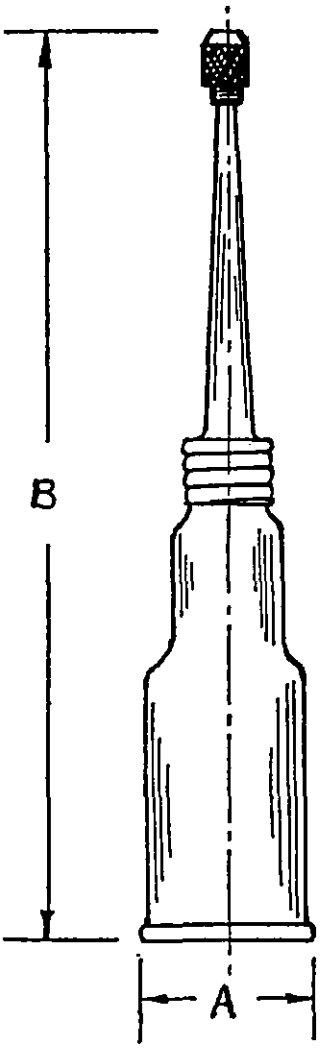


FIGURE 12. Type V, style 2, cylindrical shape, typewriter oiler.

TABLE IX. *Type V, style 2, cylindrical shape, typewriter oiler*

Size nominal	"A" diameter	"B" height	Capacity
	Minimum	Minimum	
Pint	*Inch*	*Inches*	*Fluid ounces*
½	⅛	4½	½+¼ −0

3.15.2.3 *Spout cap.* The tip end of the spout shall be provided with a threaded brass cap which shall prevent leakage when in place. The periphery of the spout cap shall be fine knurled.

3.15.3 *Style 3.* Type V, style 3, hemispherical shape oilers shall be similar to figure 13 and shall consist essentially of a body, a washer as specified in 3.4, a spout, and a spout cap as specified in 3.15.2.3. The metal shall be not less than 0.010 inch in thickness. The oiler shall be capable of discharging either one drop or several drops of oil through the tip of the spout by gently pressing the bottom of the oiler. These oilers shall comply with the requirements of table X.

3.15.3.1 *Body.* The body of the oiler shall be approximately hemispherical in shape as shown in figure 13. The opening in the top of the body shall be not less than ½ inch in diameter. The top of the body shall be threaded

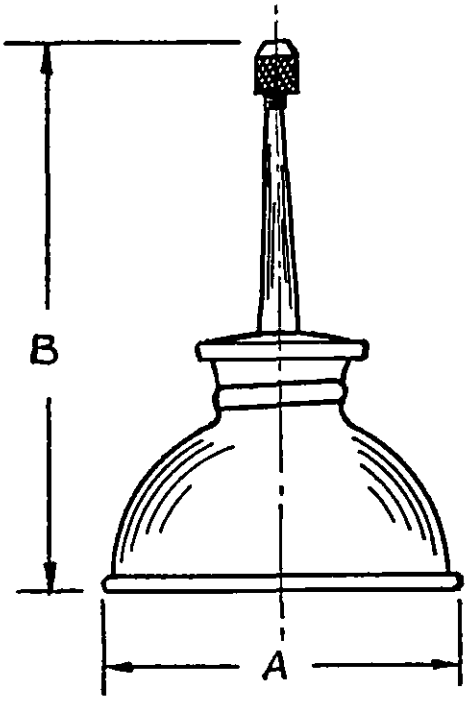


FIGURE 13. Type V, style 3, hemispherical shape, typewriter oiler.

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TABLE X. *Type V, style 3, hemispherical shape, typewriter oiler*

Size nominal	"A" diameter	"B" height	Capacity
	Minimum	Minimum	
Pint	*Inches*	*Inches*	*Fluid ounces*
½	1¾	2¾	½+¼ −0

for assembly of the spout. The bottom shall be of the push-bottom type. The bottom, when pressed, shall return to normal position when pressure is released. Failure of the bottom to return to normal position will be cause for rejection. Seams shall be leakproof. Nonmetallic filling of seams is not acceptable.

3.15.3.2 *Spout.* The spout shall conform to 3.15.2.2 except the spout shall be not less than 1¼ inches long.

3.16 Workmanship. Workmanship shall be in accordance with the best commercial practice covering this type of equipment.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance inspection.

4.2.1 *Inspection lot.* All oilers of the same type, class, style, and size offered for delivery at one time shall be considered a lot for the purposes of acceptance, inspection and tests.

4.2.2 *Sampling for examination.* A random sample of oilers from each inspection lot shall be selected in accordance with table XI, with lot acceptance based on the sampling inspection requirements in accordance with the procedures of MIL-STD-105.

TABLE XI. *Sampling for examination AQL (approx.) = 1.5 percent defective*

Number of oilers in inspection lot	Sample size, number of oilers to be taken for inspection
40 and under	Inspection level III
41 to 300	Inspection level II
301 and over	Inspection level I

4.2.3 *Sampling for tests.* A random sample of oilers shall be selected from each inspection lot in accordance with MIL-STD-105 at Inspection Level L-8, and shall be subjected to the tests specified in 4.4. The acceptable quality level shall be 2.5 percent defective. However, the sample size shall be the number associated with the letter in the table at Inspection level L-8 for each lot size and the acceptance number shall be zero until the acceptable quality level permits one or more.

4.3 Examination. Each of the sample oilers selected in accordance with 4.2.2 shall be examined to verify compliance with this specification. The sample oilers shall be checked for such points as thickness of material in the spouts, bodies, and handles. Any oiler in the sample containing one or more visual or dimensional defects shall be rejected and if the number of defective oilers in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Test procedures.

4.4.1 *Type II, class C, pump oilers with handle and plunger-operated pump.*

4.4.1.1 *Pressure-leak test.* The oilers shall

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be partially filled with oil and held with the discharge end up while the piston is moved to expel all air from the cylinder. When all air is removed, the spout shall be suitably stopped or plugged, and the piston pushed for 15 seconds with a pressure of 25 pounds applied to the handle in an attempt to expel the oil. No oil shall leak through or around the piston or at any of the joints as a result of this test.

4.4.1.2 *Capacity test.* The end of the spout shall be placed in a reservoir of oil and the piston moved through several complete strokes. The final stroke shall be such that the cylinder is empty and the piston is in position to begin the intake stroke. A moderately fast intake stroke shall be taken, and the contents of the cylinder discharged into a separate container. The amount of oil thus discharged shall be considered the capacity of the oiler. Tests shall be conducted in an atmosphere with a temperature of from 70° to 80° F. The testing media for all tests shall be No. 10 oil conforming to VV–O–526.

4.4.1.3 *Type II, classes B and C, temperature tests.*

4.4.1.3.1 *High-temperature operational test.* The filled oiler shall be heated to a temperature of 125° F. The oiler shall operate satisfactorily at this temperature.

4.4.1.3.2 *Low-temperature operational test.* The filled oiler shall be chilled to minus 40° F. The oiler shall operate satisfactorily at this temperature.

4.1.3.3 *Nonoperational test.* The empty oiler shall be heated to 160° F. and held at this temperature until saturated, then chilled to minus 80° F. The oiler shall then be allowed to assume room temperature of 70° F. The oiler shall operate satisfactorily after this temperature cycling.

4.4.2 *Type II, classes A and B.* Type II, classes A and B oilers with nylon pump parts (see 3.10.2 and 3.11.2) shall have at least one pump tested as follows: The oiler with its nylon part shall be placed into a testing device, which by cam action or by similar action against the trigger shall operate the pump at full stroke at the rate of 75 strokes per minute. The reservoir of the testing device shall be filled with lubricating oil which shall be pumped through the oiler and returned to the reservoir by tubing attached to the spout of the oiler. The pump shall be actuated at least one million full strokes and after this test shall operate efficiently in its intended use. A statement signed by a responsible representative of the manufacturer or a signed test report from a recognized testing laboratory that the oilers have passed this test will be accepted as proof of compliance, provided the oilers meet all other applicable tests herein, including the tests in 4.4.1 and 4.4.4.

4.4.3 *Type V, typewriter oilers.* The testing of this type oiler will be limited to operational test to determine compliance with 3.15.1, 3.15.2, or 3.15.3 as applicable.

4.4.4 *All other types of oilers.*

4.4.4.1 *Pressure test.* The sample oilers of welded, brazed or silver soldered joint construction selected for tests shall be subjected to a hydrostatic pressure test of 5 pounds per square inch for a period of 5 minutes and shall show no evidence of leaking at the seams, joints, threads or other connections. Oilers of lead-tin solder joint construction shall be tested for leakage but no hydrostatic pressure tests will be required.

4.4.4.2 *Spout load test.* The oilers shall be subjected to the following spout load test:

(*a*) Assemble the spout tightly to the body.

(*b*) Support the body of the oiler with the spout extended in a horizontal plane.

(*c*) Apply a 75-inch-pound load within 1½ inches of the vent end of the spout; as an example, apply a 10-pound load 7½ inches out on the unsupported end of the spout of an oiler with a 9-inch spout. In lieu of using a suspended weight, the same test can be accomplished by holding the oiler body in the hands and applying the pressure on a platform scale with the predetermined load set up on the scale. The seam between the spout cap and the spout or any other seam shall not be opened up

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after this test and the spout shall not be bent nor show any other signs of permanent deformation.

4.4.4.3 *Test of welded seams in the spout construction.* A spout from one of the welded seam construction oilers shall be removed and hammered flat for one-half of its length with the seam at the bend, and for the remainder of its length with the bend 90 degrees from the seam, and under this test shall show no evidence of opening of the seam.

4.4.4.4 *Test of brazed or soldered seams in the spout construction.* A brazed or silver soldered seam type of spout shall be removed from one of the oilers and one 90-degree bend made with the seam in the middle of the bend, and under this test shall show no evidence of opening of the seams.

4.4.4.5 *Welded filler spout test.* A welded seam type of spout on one of the fillers shall be hammered, as described in 4.4.4.3 without removal from the filler and shall not open at the seam or separate from the body.

4.4.4.6 *Filler handle test.* With the handle and bottom rim of the filler supported on an anvil, the handle shall be hammered from the top side until the top and bottom part of the handle are brought within ½ inch of each other at 1 inch from the bend. After this test, neither welded joint of the handle to the body shall show evidence of opening.

4.4.4.7 *Further uses of samples.* The samples used in the spout tests shall be used for checking the thickness of the material in spout, handles, bottom and the body, as specified herein. These samples shall also be used to check the seam, joints, and also the thickness of material at the root of machine-cut threads which shall be at least equal to the thickness of the oiler body. The thickness of material at the root of the threads may include more than one thickness of material provided the layers are welded or brazed into one component.

4.5 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the oilers shall be examined and tested to determine conformance with section 5. Examination of packaging requirements not covered by referenced specifications shall be performed in accordance with MIL-P-116. Examination of packing and marking requirements not covered by referenced specifications shall be in accordance with MIL-STD-105, using an AQL of 4.0 percent defective.

5. PREPARATION FOR DELIVERY

(Civil agencies. Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Preservation and packaging (see 6.2).

5.1.1 *Level A.* Oilers shall be preserved and packaged in accordance with method III of MIL-P-116. Contact preservative shall not be required. Quantity per unit package shall be as specified in the contract or order. Unit containers shall be folding cartons or set-up boxes conforming to PPP-B-566, PPP-B-676, PPP-B-665, or PPP-B-636 at the option of the contractor. Box closure shall be in accordance with the applicable box specification or appendix thereto.

5.1.1.1 *Intermediate containers.* Unit quantities in an intermediate container shall be as specified (see 6.2). Intermediate containers shall conform to PPP-B-566, PPP-B-676, PPP-B-665, or PPP-B-636, at the option of the manufacturer. Box closures shall conform to the applicable box specification or the appendix thereto. The gross weight of paperboard boxes shall not exceed 10 pounds and fiber boxes shall not exceed 20 pounds.

5.1.2 *Level C.* Oilers shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing (see 6.2).

5.2.1 *Level A.* Hand oilers, of one type, class, and style packaged as specified (see 6.2) shall be packed in overseas type, wood-cleated fiberboard, nailed wood, fiber, wire-bound wood, wood cleated veneer paper overlaid, or wood cleated plywood boxes conforming to PPP-B-591, PPP-B-621, PPP-B-636 (class 2), PPP-B-585 (class 3), PPP-B-576, or PPP-B-601, respectively at

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the option of the manufacturer. Unless otherwise specified, shipping containers shall have caseliners conforming to MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Caseliners for boxes conforming to PPP-B-636 may be omitted provided all joints and corners of the boxes are sealed with minimum 1½-inch-wide tape conforming to PPP-T-76. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto, except fiber boxes may be reinforced with pressure-sensitive tape conforming to type IV of PPP-T-97 and the appendix thereto. The gross weight of wood or wood cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification. Intermediate fiber boxes conforming to class 2 or 3 of PPP-B-636, closed, sealed and banded as specified herein, and used as the shipping container need not be overpacked.

5.2.2 *Level B.* Hand oilers, of one type, class, and style, packaged as specified (see 6.2), shall be packed in domestic type wood cleated fiberboard, nailed wood, wirebound wood, cleated plywood or wood cleated veneer paper overlaid boxes or class 2 fiber boxes conforming to PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-601, MIL-B-10377 or PPP-B-636, respectively at the option of the contractor. Box closure shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification. Intermediate fiber boxes conforming to PPP-B-636, closed as specified herein, and used as the shipping container need not be overpacked.

5.2.3 *Level C.* Hand oilers, packaged as specified (see 6.2), shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with rules or regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Military marking.* In addition to any special marking required by the contract or order, unit and intermediate packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 *Civil agencies.* In addition to any special marking specified in the contract or order, each unit and intermediate package shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. Oilers are also known as "oil cans", "pistol oilers", "fillers", "trigger oilers", "hydraulic oilers", and other similar designations, dependent upon design and construction. Pistol and trigger oilers are classified in this specification as pump oilers without handles. Oilers might be used for lubricating machinery or automotive equipment with any type of oil that will flow, similar to 8, 10, 20 or 30 grades of oil covered in VV-O-526, except that pump oilers without handles are generally used for lighter oils such as grades 8 or 10. The filler type is generally used for filling large oil cups on heavy machinery with the heavier oils such as 20 or 30 grade.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class and style required (see 1.2).
(*c*) Size required (see applicable table).
(*d*) If flexible spouts are required (see 3.3).
(*e*) Length of spout (see applicable type).
(*f*) Copper plating method, if different (see 3.6.2).
(*g*) Corrosion-resistant metals, coating or plating, if different.
(*h*) Type of spout required (see 3.8.2, 3.10.4 and 3.11.3).
(*i*) Whether adapter and spray nozzle are required (see 3.11.3).

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(*j*) Type of thread required for type II, class C oiler (see 3.12.2).
(*k*) If more than one spout is required, state the additional style(s) of spout(s) required (see 3.12.6).
(*l*) Whether the oilers shall be so designed as to function between minus 40° to plus 125° F. and withstand storage temperatures between minus 80° to plus 160° F. without harmful effects (see 3.12.7).
(*m*) Whether the type III filler shall be made of copper or steel (see 3.13).
(*n*) If a specific style (shape) of type V typewriter oiler is required (see 3.15).
(*o*) Applicable level of preservation, packaging and packing (see 5.1 and 5.2).
(*p*) Unit quantities in intermediate containers (see 5.1.1.1).
(*q*) Marking, if different (see 5.3).

6.3 Items procured under this specification for Military use are to be limited to the variety shown on the applicable Military Standards. Personnel of the Military departments are requested to refer to these documents for guidance.

6.4 Supersession data. This specification supersedes Interim Federal Specification GGG–O–591d (GSA–FSS) dated February 13, 1962 and Federal Specifications GGG–O–591a dated May 5, 1954, RR–C–87 dated December 19, 1938, and RR–O–376a dated April 30, 1953. This specification includes the requirements of Military Specifications MIL–O–45029(Ord) dated January 22, 1958 and MIL–G–3882a, dated March 7, 1960.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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